Work & Energy

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Q1. A body of mass 1 kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from x = 0 to 5 m is _____ J.

JEE Main 2026 (Online) 6th April Evening Shift

✓ Answer: 1250

Solution:

By the work–energy theorem, $W = \Delta KE$.

At x = 5 m: $v = 2(5)^2 = 50$ m/s.

$KE_{final} = \frac{1}{2}mv^2 = \frac{1}{2}(1)(50)^2 = 1250$ J.

$KE_{initial} = 0$ (since v = 0 at x = 0).

So $W = 1250 - 0 = 1250$ J.

Q2. A rain drop of mass 1 g starts with zero velocity from a height of 1 km. It hits the ground with a speed of 5 m/s. The work done by the unknown resistive force is _____ J. (Take $g = 10$ m/s²)

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✓ Answer: −9.9875

Solution:

Work done by gravity: $W_g = mgh = (0.001)(10)(1000) = 10$ J.

Change in kinetic energy: $\Delta KE = \frac{1}{2}mv^2 - 0 = \frac{1}{2}(0.001)(5)^2 = 0.0125$ J.

By the work–energy theorem: $W_g + W_{resistive} = \Delta KE$.

$W_{resistive} = 0.0125 - 10 = -9.9875$ J (the resistive force removes almost all of the gravitational work as the drop falls).

Source: JEE Main 2026 (NTA official releases, physics section) · formulaverse.in